Combination circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Reduce a circuit by identifying genuine series and parallel groups.

SUCCESS CRITERIA - I can...

1. Identify nodes before grouping resistors.
2. Reduce one group at a time.
3. Check current and voltage conservation in the original circuit.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “parallel circuit” to explain your model or result. Sentence starters are optional.

series: connected along one unbranched current path

parallel circuit: branches connected across the same two nodes

node: a shared electrical connection point

01 Warm-up: bring back an earlier idea

A 3 Ω resistor carries 5 A.
Find the potential difference across it.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

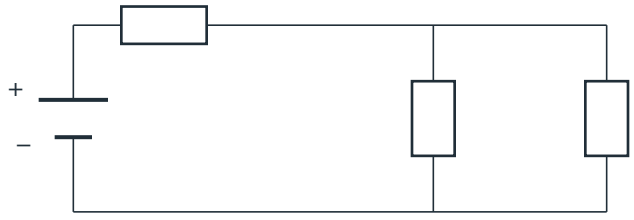
02 Worked example: explain the model

Reference: $R_{series} = R_1 + R_2$; $1/R_{parallel} = 1/R_1 + 1/R_2$; $V = IR$. Reduce one known group at a time. Check nodes before calling components series or parallel.

Worked reasoning: A 2 ohm resistor is in series with two 4 ohm resistors in parallel. The parallel pair equals 2 ohms; total 4 ohms. A 12 V supply gives 3 A total.

Your turn: Draw the network and label the two series sections before splitting the branch current.

Sentence starter: The … in my model represents … . Its label shows … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A 2 Ω resistor is in series with two parallel 6 Ω resistors.
Draw the arrangement and circle the parallel group.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A 10 V battery powers the network in question 3: a 2 Ω resistor in series with two parallel 6 Ω resistors. You may use the supplied parallel equivalent, 3 Ω.
Label each section, find total resistance and calculate source current.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student adds all three resistor values because they are drawn side by side. What should determine the grouping instead?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A radar echo’s complete outward-and-return path is 6000 m.
Find the target range and explain the division.
Label the outgoing and returning paths.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

If 2 A reaches two equal parallel branches, find each branch current.
Label the circuit and show the current path or paths.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Replace the parallel pair by two 8 ohm resistors.
Find the new equivalent resistance and total current on 12 V.
Label the circuit and show the current path or paths.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3